Mittuniversitetet
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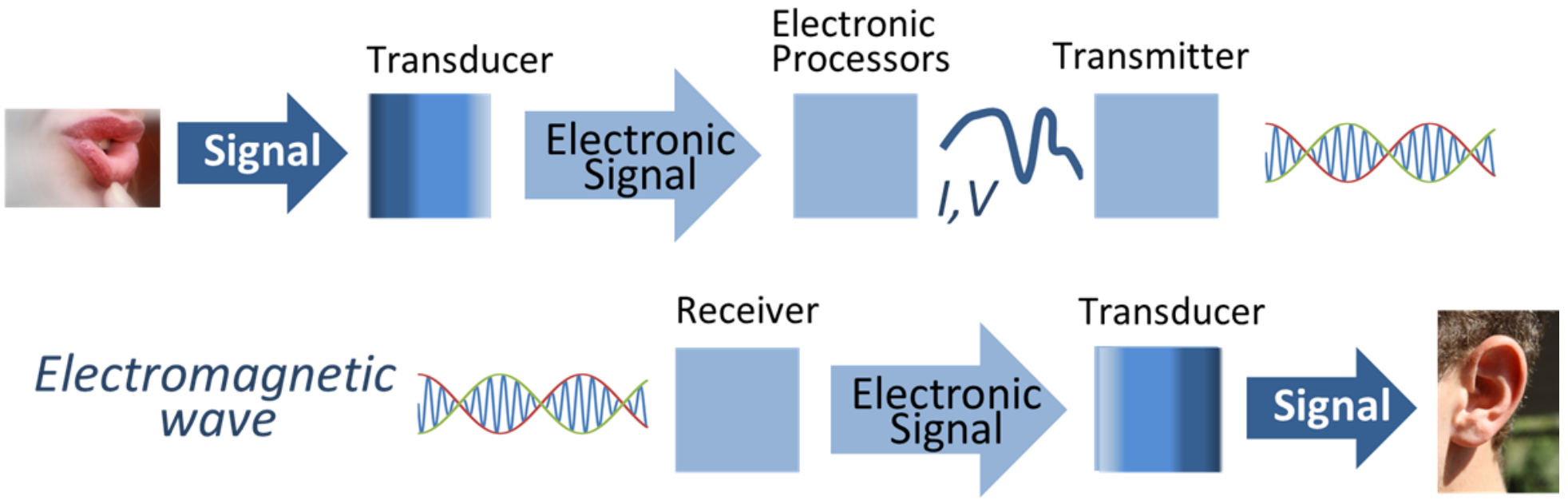
DT081A – Signal and Image Processing

Lecture 2 – Sampling, frequency domain, and convolution



Roger Olsson (PhD)
Department of Computer and Electrical Engineering

Signals are everywhere



Sensor to acquire the signal



Audio sensor

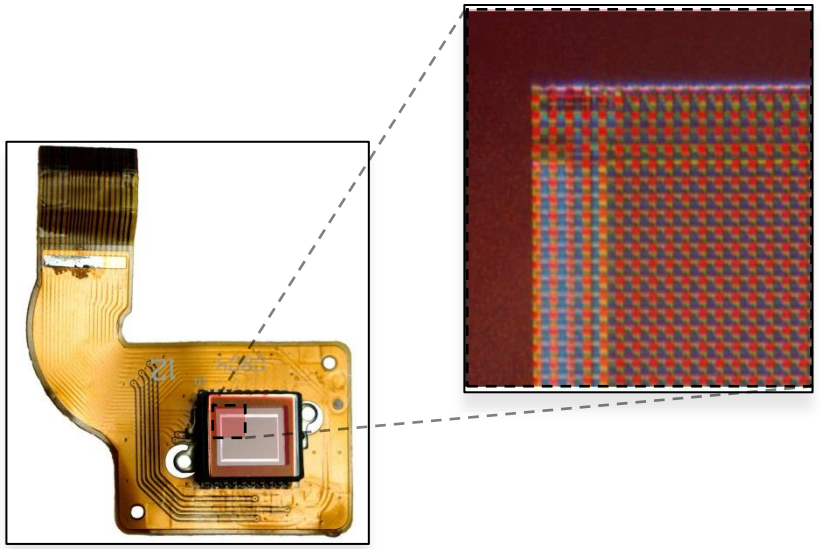
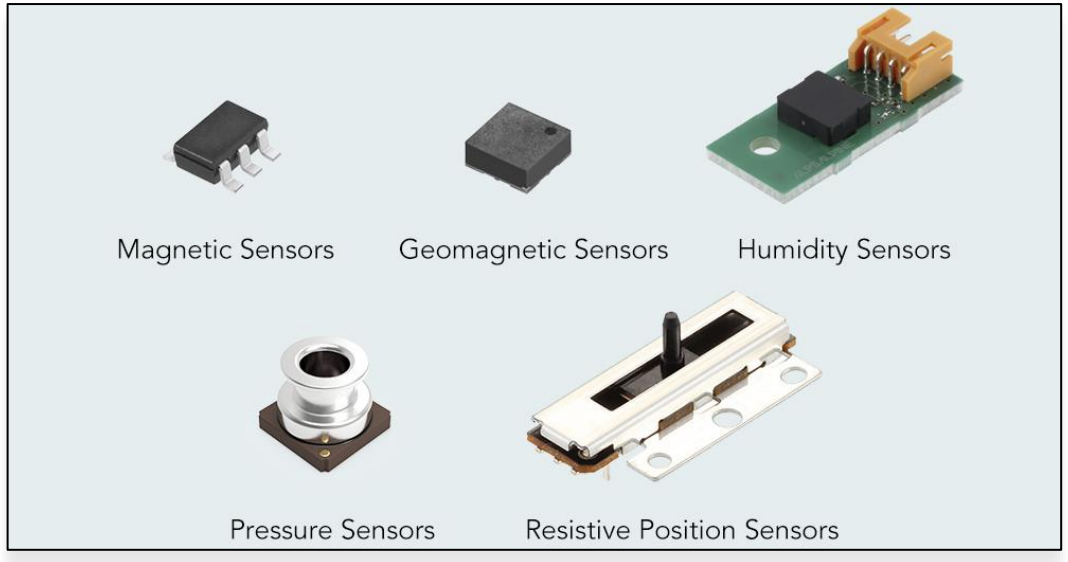
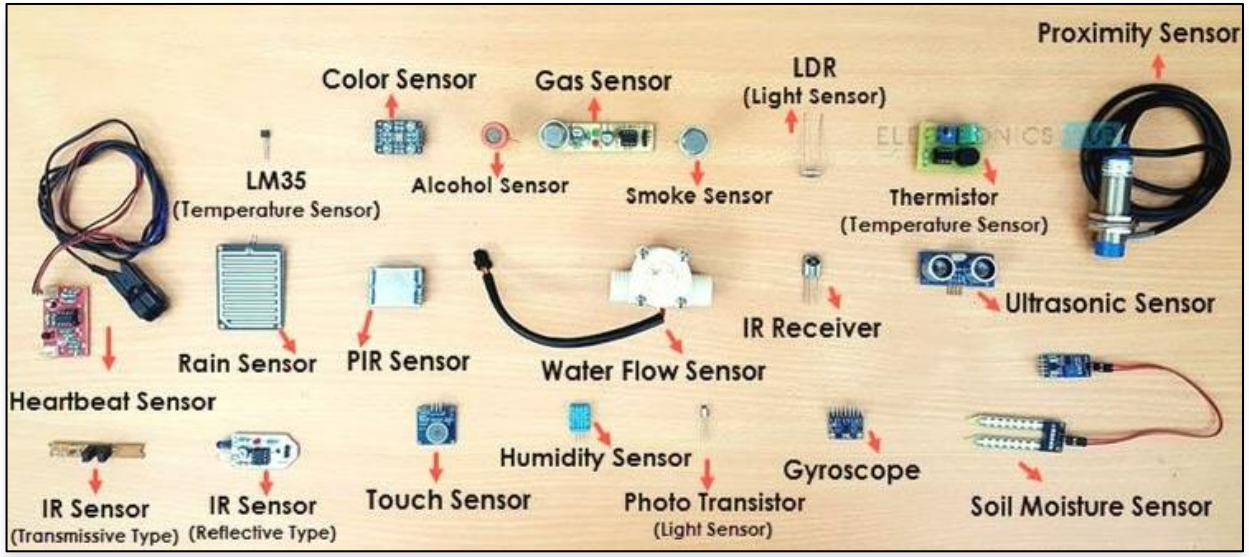


Image sensor

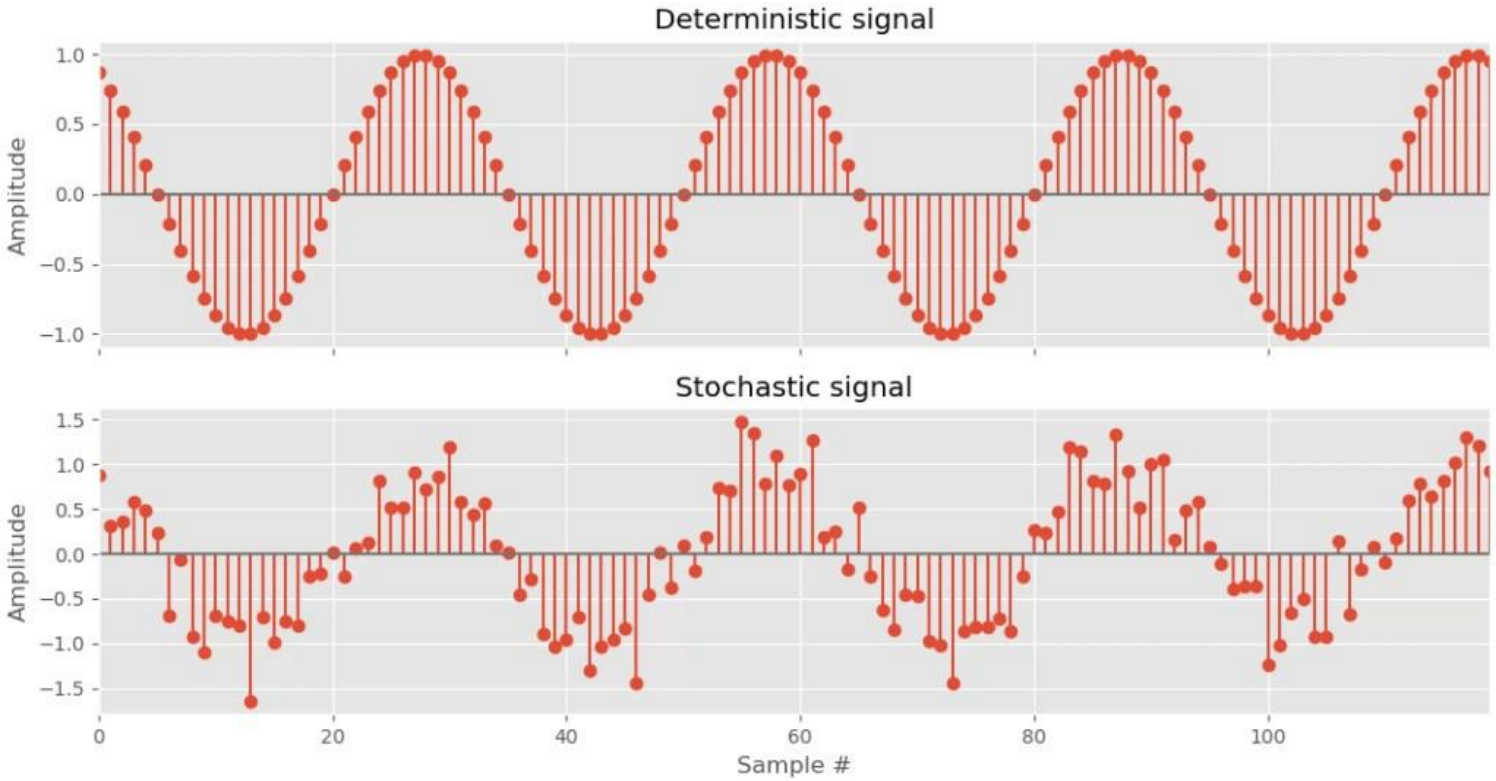


Sensors from Alps Alpine. Source: Alps Alpine



Different Types of Sensors. Source: electronicshub

Deterministic vs stochastic signals



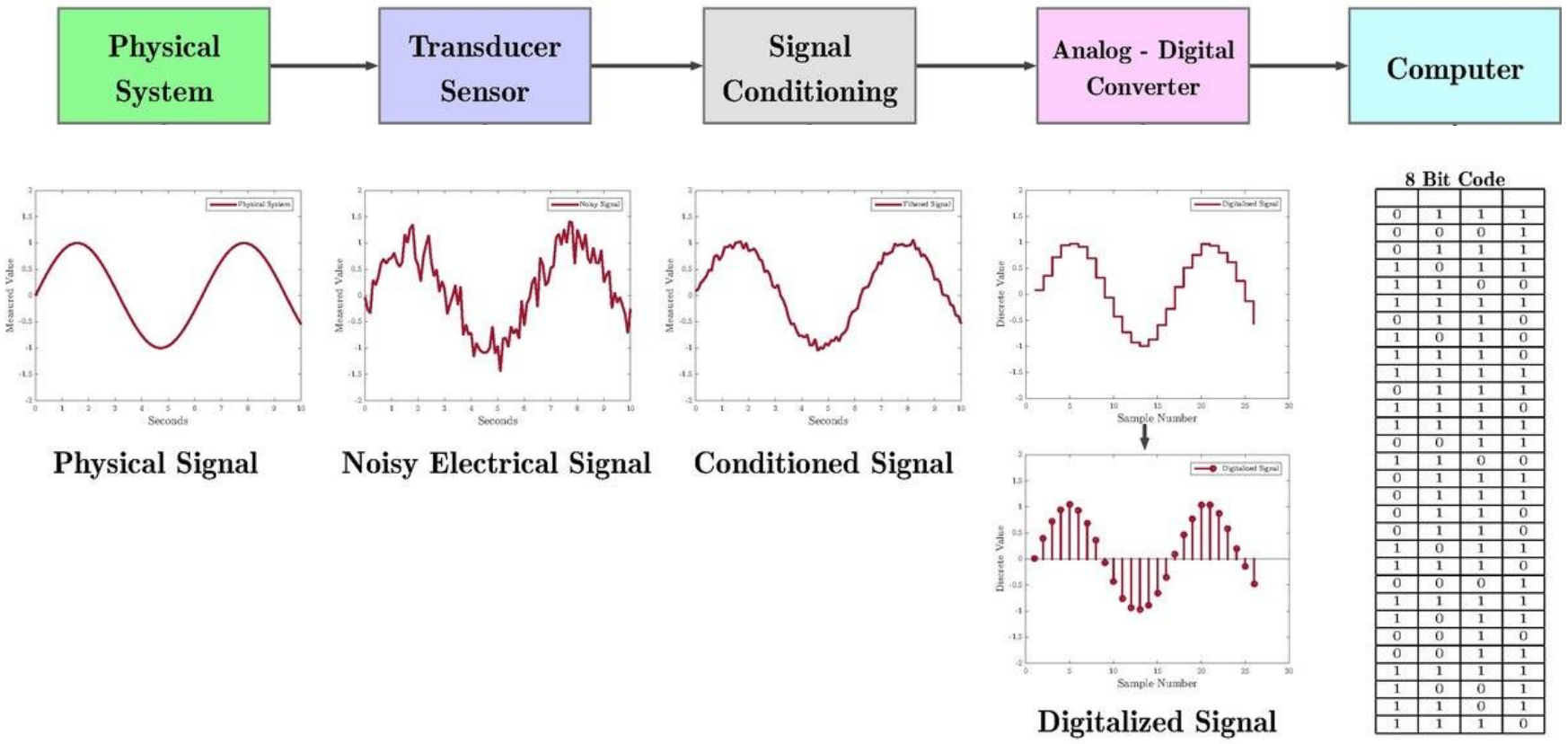
$$y(n) = \sin\left(2\pi \frac{f}{f_s} n + \varphi\right)$$

+ Predictable, completely described by mathematical function
+ Easy to analyze
- Not realistic

$$z(n) = y(n) + w(n), \quad w(n) \sim \mathcal{N}(0, \sigma^2)$$

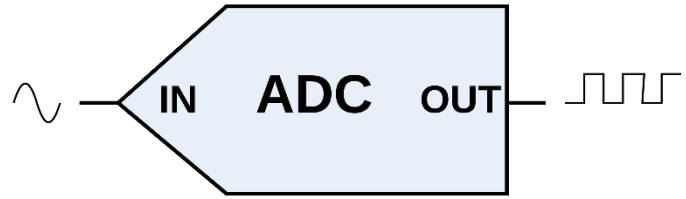
+ Unpredictable, modelled using probability distribution functions
+ Real-world signals
- More complicated analysis

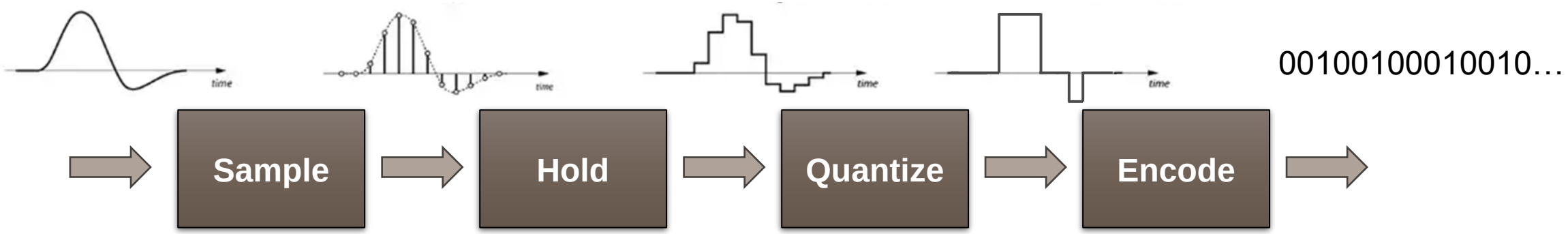
Overview of the signal acquisition process



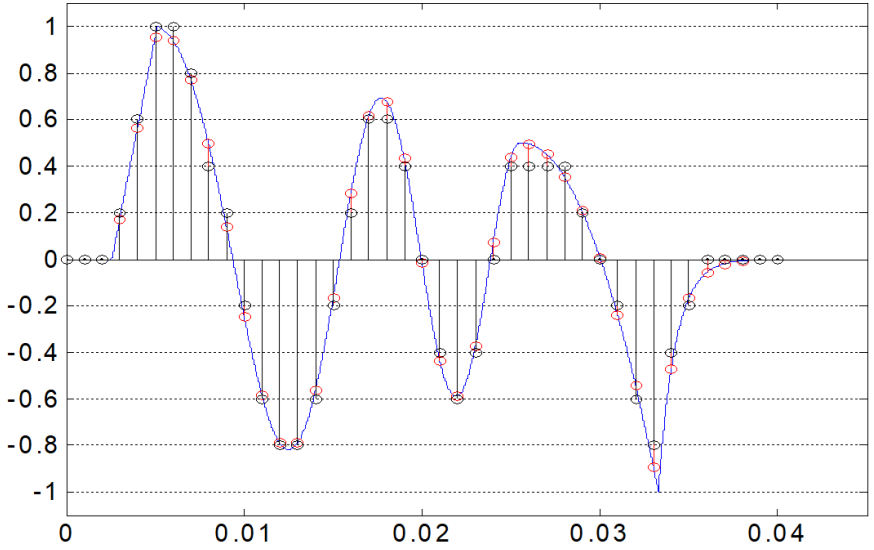
https://en.wikipedia.org/wiki/Data_acquisition

Analog-to-Digital Conversion (ADC) components





Continuous to discrete conversion



- Recap time-discrete signal $x(n)$

  $f: \Omega \to \mathbb{R}, where\ \Omega = \{n \quad | \quad a \leq n \leq b\} \subseteq \mathbb{Z}$

  $s = f\left(n\right) \in \mathbb{R}, n \in \left[a, b\right] \subseteq \mathbb{Z}$

- Continuous to Discrete Conversion

  Continuous-time signal

  $$x_c(t),\ where\ t \in \mathbb{R}$$

  Sampling period

  $$T\ (in\ seconds)$$

  Sampling process

  $$f(n) = x_c(nT), n \in \mathbb{Z}$$

Continuous to discrete conversion

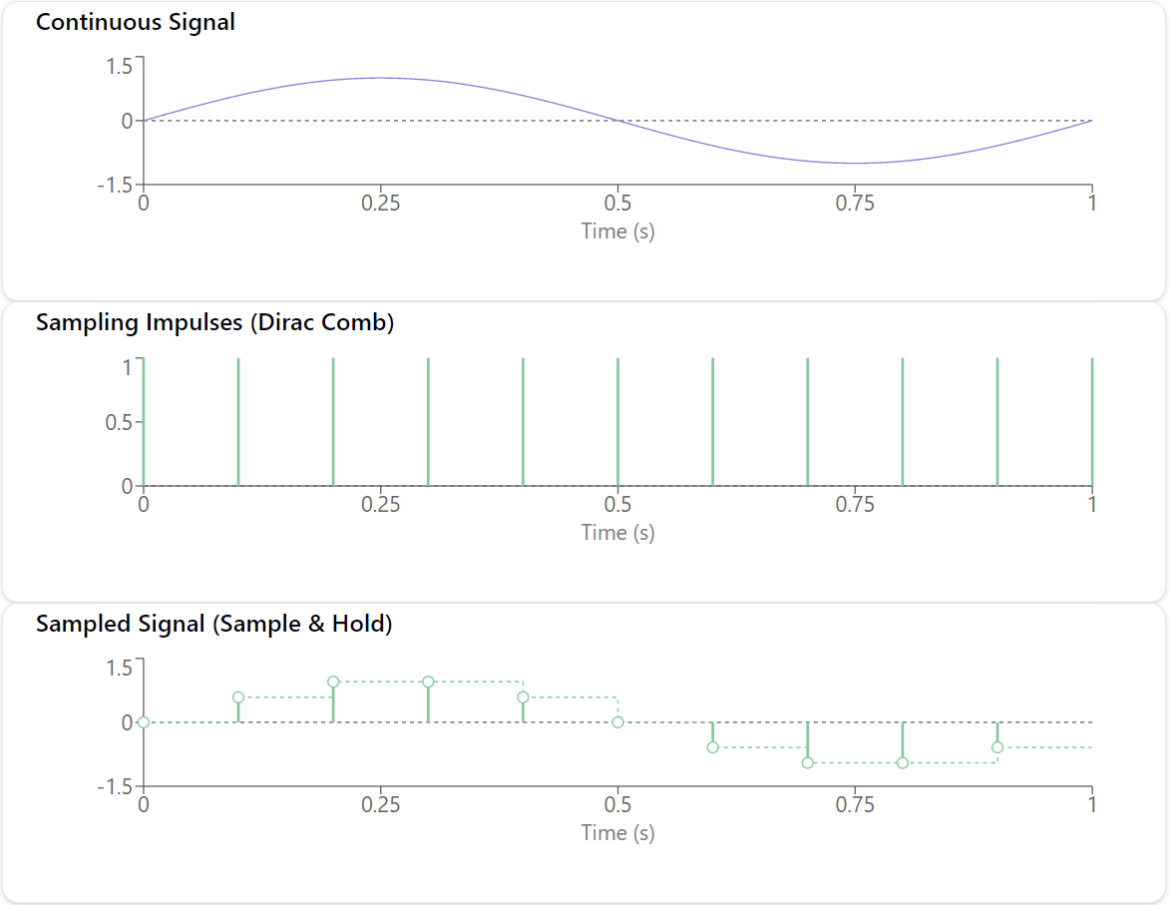
- From Continuous to Discrete Domain

  Continuous sampled signal

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT)$$

  Discrete representation

$$s = f(n) = x_c(nT), n \in \Omega \subseteq Z$$



Demo

Fourier transform expanded

- Continuous Fourier Transform (CFT)

$$X(f) = \mathcal{F}x(t) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\,dt$$

- Inverse CFT

$$x(t) = \mathcal{F}^{-1}X(f) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}\,df = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,d\omega$$

- Discrete-Time Fourier Transform (DTFT)

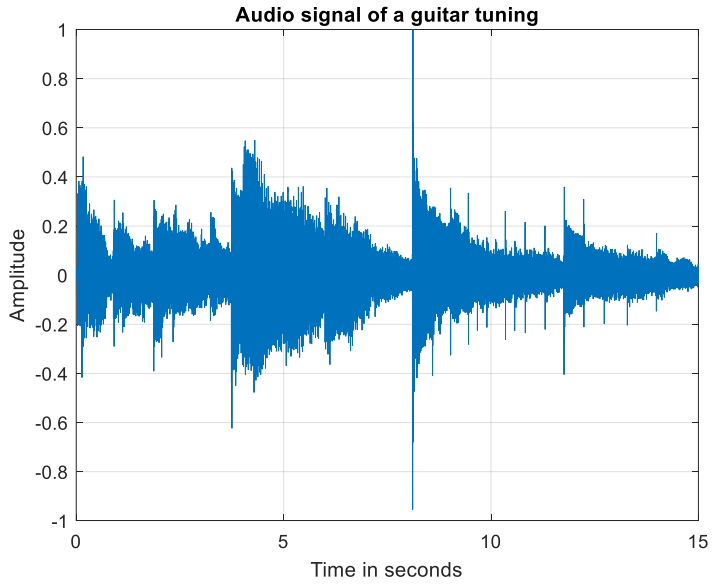
$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

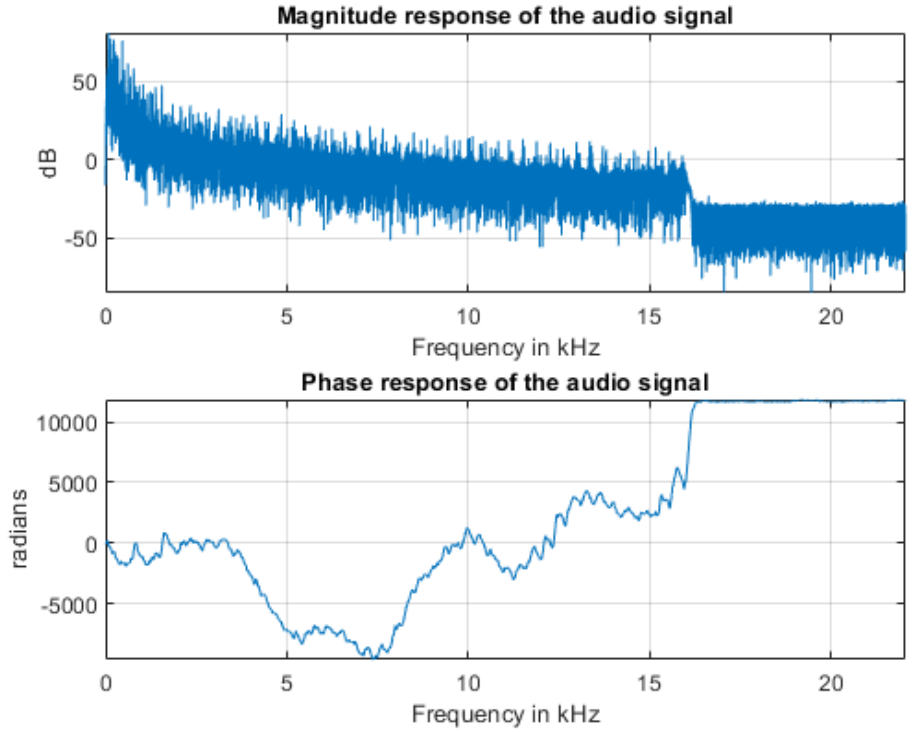
- Inverse DTFT

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega n}\,d\Omega$$

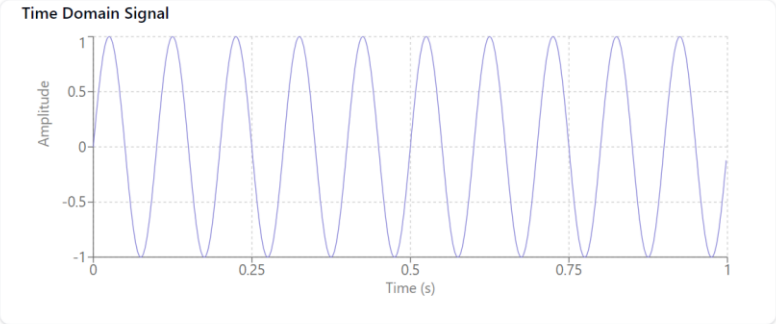
Fourier transform – frequency domain representation

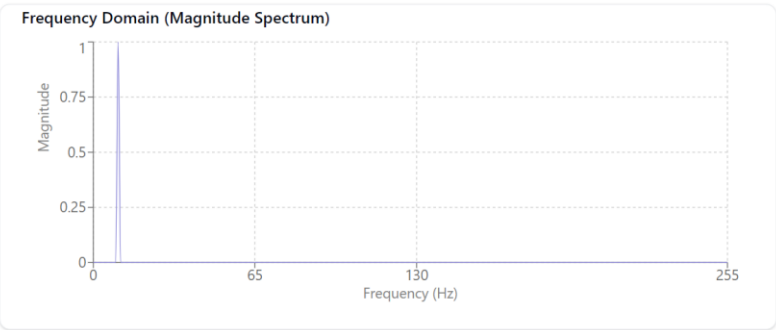
- Magnitude $|X(f)|$ Strength of each frequency component
- Phase $\angle X(f)$ Time delay of each frequency component

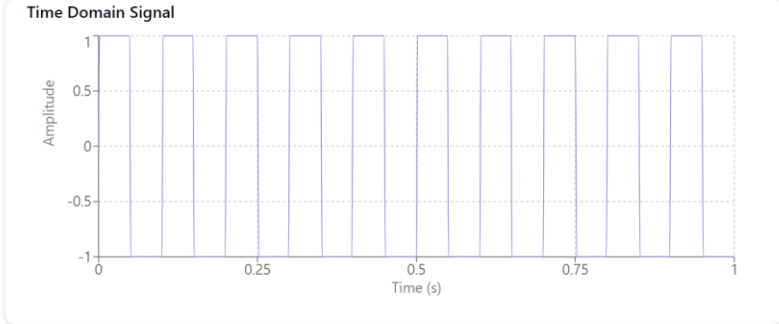


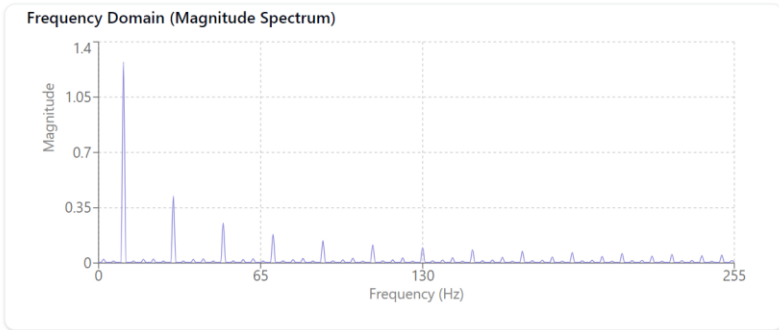


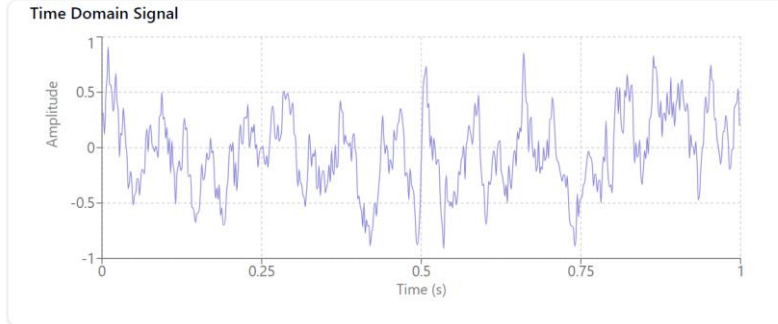
Signal examples in time and frequency domain

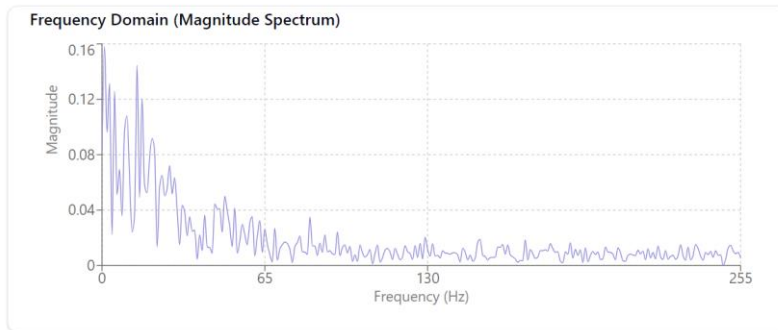












Demo

Fourier transform properties

- Linearity

$$\mathcal{F}\{ax(t) + by(t)\} = aX(f) + bY(f)$$

- Time-shifting

$$\mathcal{F}\{x(t - t_0)\} = X(f)e^{-j2\pi f t_0}$$

- Frequency-shifting

$$\mathcal{F}\left\{x(t)e^{j2\pi f_0 t}\right\} = X(f - f_0)$$

- Convolution (*)

$$\mathcal{F}\{x(t) * y(t)\} = X(f)Y(f)$$

$$\mathcal{F}\{x(t)y(t)\} = X(f) * Y(f)$$

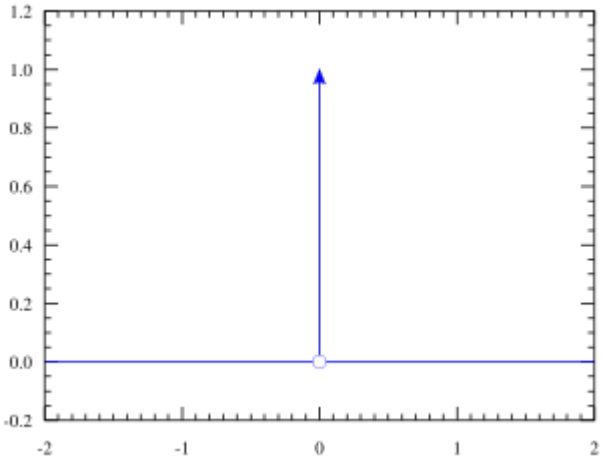
Continuous to discrete conversion

- Mathematical Representation of Sampling

  Dirac function

$$\delta(x) \simeq \begin{cases} \infty, x = 0 \\ 0, x \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x)\ dx = 1$$



Schematic representation of the Dirac delta function by a line surmounted by an arrow.

  Sampling function

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

  Sampled signal

$$x_s(t) = x_c(t) \cdot \delta_T(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT)$$

Convolution derivation

Remember

$x(t) \xrightarrow{\mathcal{F}} X(f)$

$y(t) \xrightarrow{\mathcal{F}} Y(f)$

$z(t) \xrightarrow{\mathcal{F}} Z(f)$

- Given two signals $x(t)$ and $y(t)$ form their product $z(t) = x(t) \cdot y(t)$
- Express $Z(f)$ using the Fourier transform $Z(f) = \mathcal{F}z(t) = \int_{-\infty}^{\infty} z(t)e^{-j2\pi ft}\,dt$
- Substitute $z(t) = x(t)y(t)$ $Z(f) = \int_{-\infty}^{\infty} x(t)y(t)e^{-j2\pi ft}\,dt$
- Express $y(t)$ using its inverse Fourier transform $y(t) = \int_{-\infty}^{\infty} Y(\tau)e^{j2\pi\tau t}\,d\tau$
- Substitute this into the equation for $Z(f)$ $Z(f) = \int_{-\infty}^{\infty} x(t)\left[\int_{-\infty}^{\infty} Y(\tau)e^{j2\pi\tau t}\,d\tau\right]e^{-j2\pi ft}\,dt$
- Rearrange the integrals $Z(f) = \int_{-\infty}^{\infty} Y(\tau)\left[\int_{-\infty}^{\infty} x(t)e^{-j2\pi(f-\tau)t}\,dt\right]d\tau$
- Recognize the inner integral as the Fourier transform of $x(t)$ $Z(f) = \int_{-\infty}^{\infty} Y(\tau)X(f-\tau)d\tau$

Convolution derivation

- Key Result and Fourier Transform Property

$$Z(f) = X(f) * Y(f) = \int_{-\infty}^{\infty} X(\tau)Y(f-\tau)d\tau$$

Multiplication in time domain ⇔ Convolution in frequency domain

- Application to Sampling

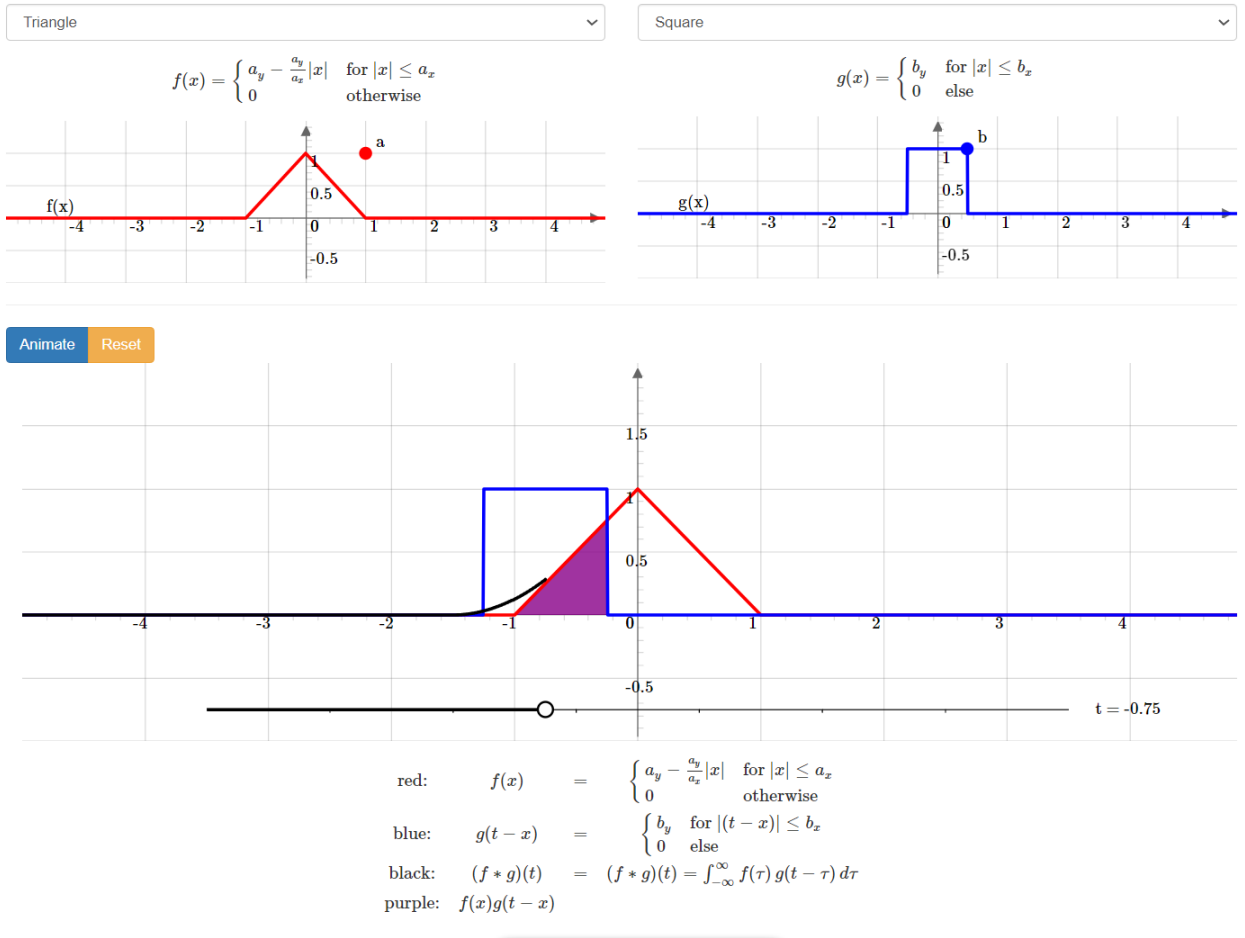
Time domain $x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$

Frequency domain $X_s(f) = X(f) * \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$ where $f_s = 1/T_s$ is the sampling frequency

What to expect of the sampled signal's frequency domain representation?

Spectral replication of the continous spectrum

Convolution



Demo

Source: [phiresky](#)'s public GitHub repositories

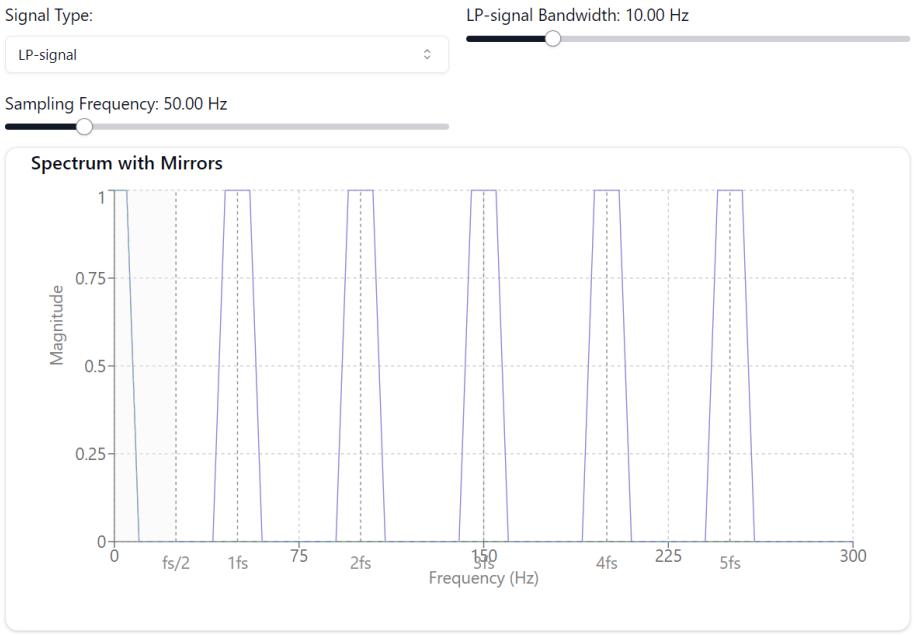
Application to sampling

Time domain $x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$

Frequency domain $X_s(f) = X(f) * \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$ where $f_s = 1/T_s$ is the sampling frequency

What to expect of the sampled signal's frequency domain representation?

- Spectral replication of the continous spectrum



Demo

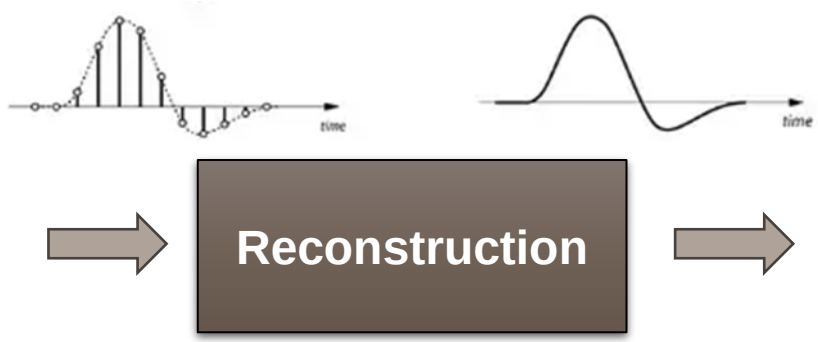
Reconstruction I

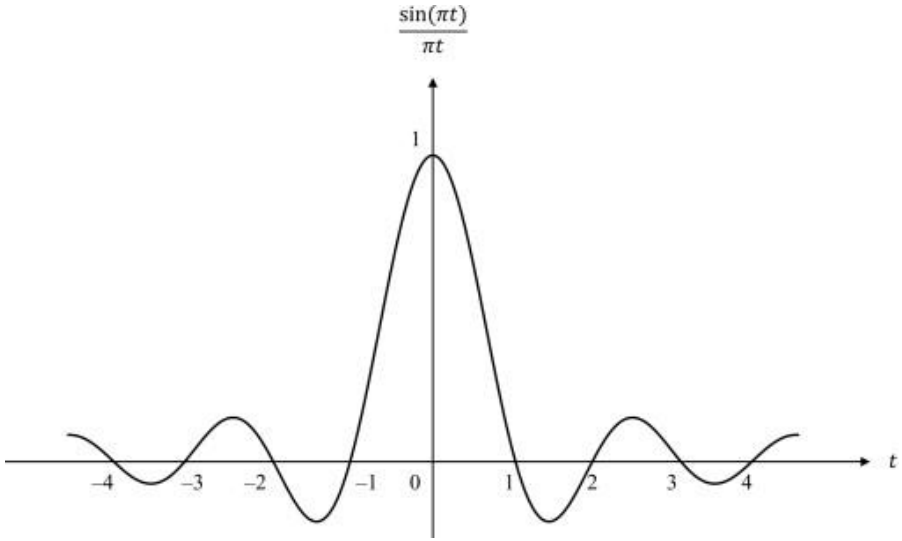
Time Domain Perspective

Recover continuous-time signal
$x(t)$ from its samples $x[n] = x(nT_s)$

Ideal Reconstruction Formula

$x(t) = \sum_{n=-\infty}^{\infty} x[n]\text{sinc}\left(\frac{t-nT_s}{T_s}\right)$, where $\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$?



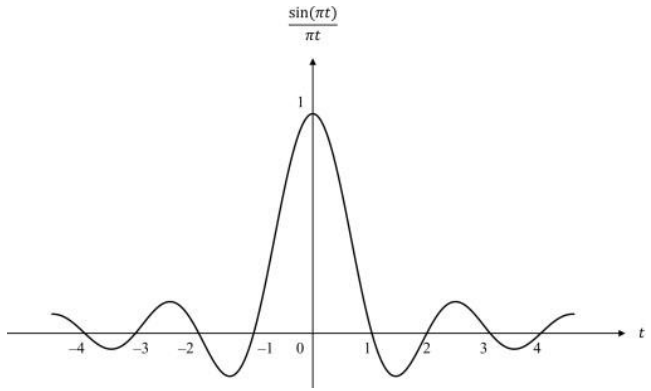


Reconstruction II

Frequency Domain Perspective

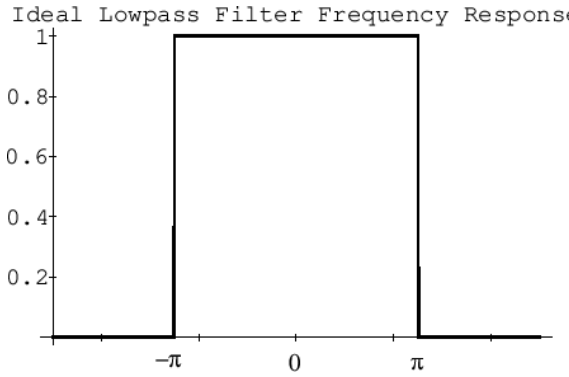
Sampled signal spectrum

$$X_s(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(f - kf_s)$$



$\mathcal{F}$

$\mathcal{F}^{-1}$



Reconstruction filter

$$H(f) = T_s rect(fT_s) = \begin{cases} T_s & |f| < \frac{1}{2T_s} \\ 0 & otherwise \end{cases}$$

Multiply sampled signal spectrum with ideal lowpass filter / reconstruction filter
$x_r(t) = x_s(t) * h(t)$,
where $h(t) = \text{sinc}(t/T_s)$

Reconstruction III

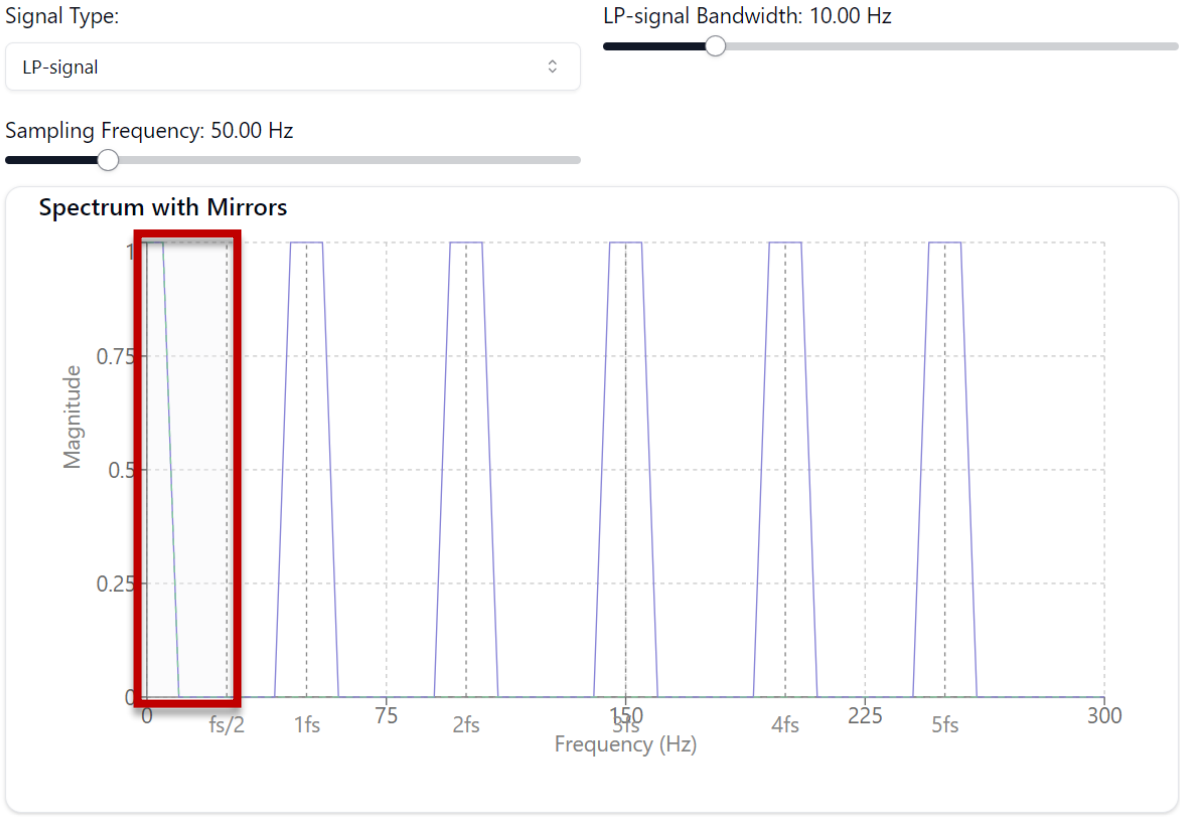
Frequency Domain Perspective

Sampled signal spectrum

$$X_s(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(f - kf_s)$$

Reconstruction filter

$$H(f) = T_s rect(fT_s) = \begin{cases} T_s & |f| < \frac{1}{2T_s} \\ 0 & otherwise \end{cases}$$



Reconstruction Process

Multiply sampled signal spectrum with ideal lowpass filter / reconstruction filter

$x_r(t) = x_s(t) * h(t)$, where $h(t) = \text{sinc}(t/T_s)$

Demo

Nyquist-Shannon Sampling Theorem

Perfect reconstruction is possible if $f_s > 2f_{\max}$

$f_{\max}$ is the highest frequency component in $x(t)$

Practical Considerations

Ideal sinc interpolation is not realizable (infinite duration)

Practical reconstruction uses approximations (e.g., zero-order hold, linear interpolation)

Key Takeaway
Reconstruction recovers the original continuous signal from its samples by filtering out the spectral replicas introduced by sampling.

Aliasing – causes, effects, and prevention

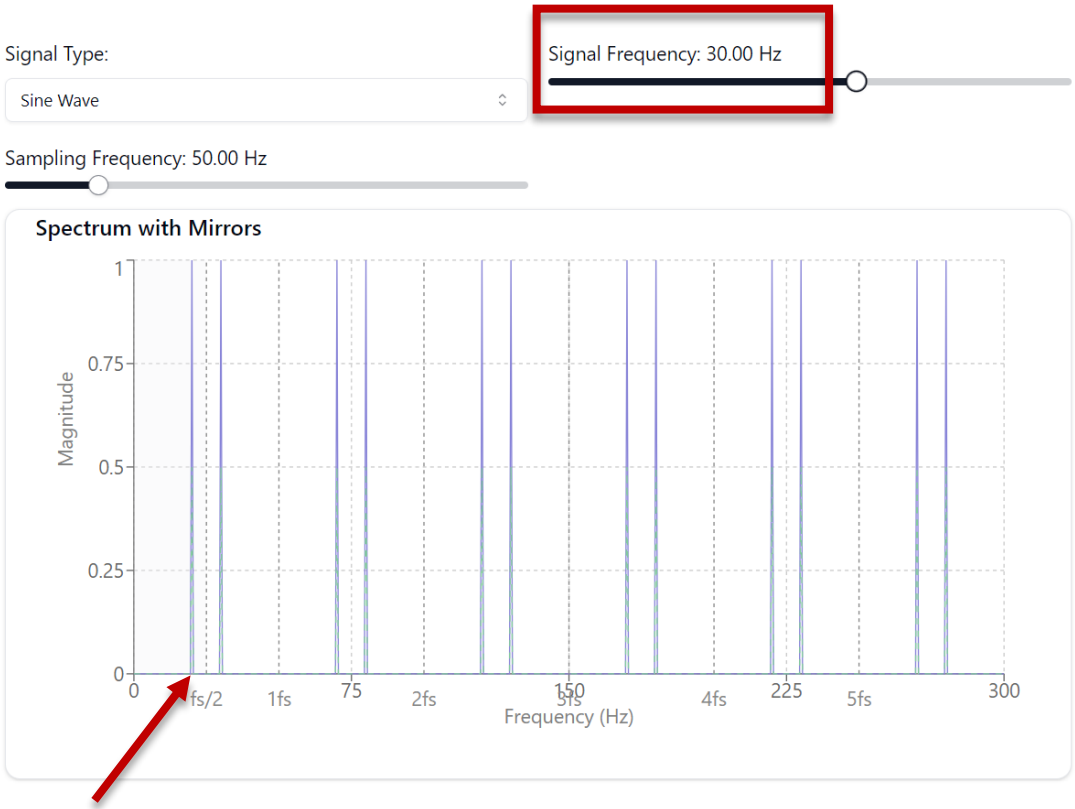
Occurs when $f_s < 2f_{max}$,
i.e. signal is sampled at a rate insufficient to capture its highest frequency components

Undersampling – when sampling rate is too low for signal's bandwidth

Spectral overlap in frequency domain

New frequencies are reconstructed

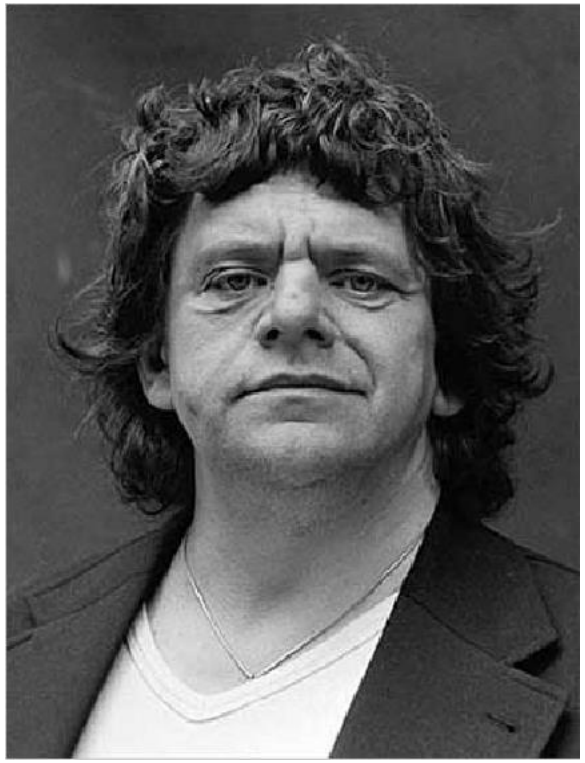
$f_{alias} = |f - nf_s|$, where $n \in \mathbb{Z}$



$f_{alias} = |30 - 1 \cdot 50| = 20Hz$

Demo

Image sampling



$f_s \cong 2f_{max}$



$\frac{f_s}{2}$



$\frac{f_s}{4}$

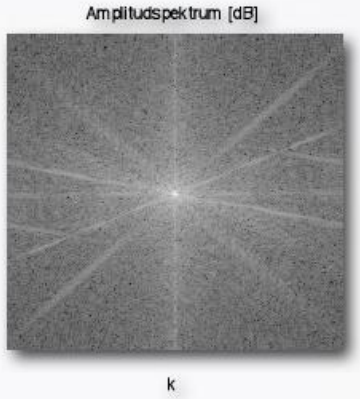


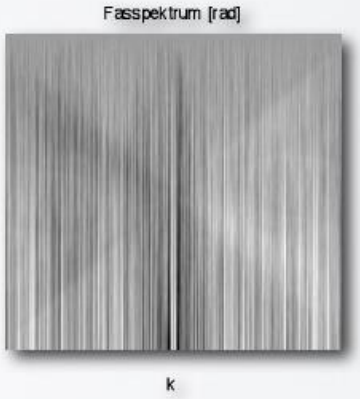
$\frac{f_s}{10}$

Spatial Frequency

- Spatial frequency = the rate of pixel value change over a specific length (# of pixels, mm, inches etc.)
- The frequency representation of an image consists of
  - Amplitude spectrum
  - Phase spectrum

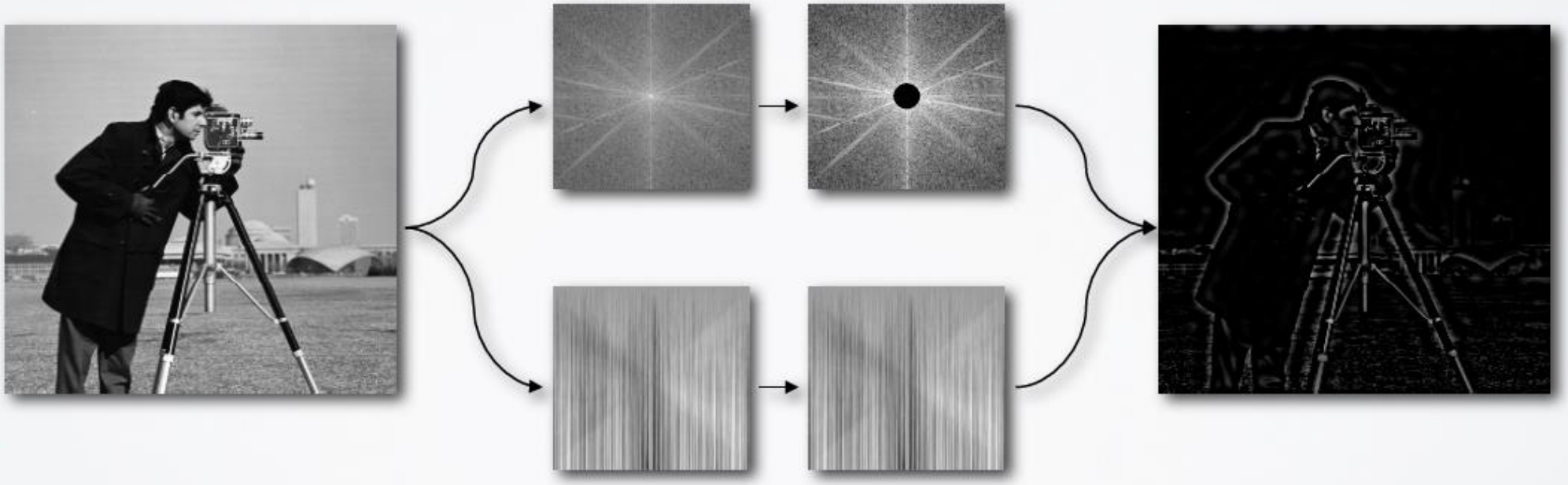






Spatial Frequency Domain

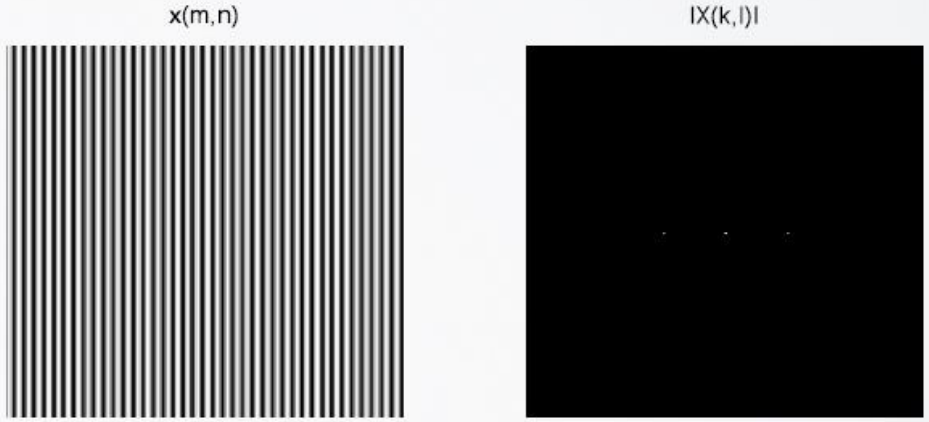
- Processing can be conducted in the spatial frequency domain as well as in the pixel domain of the original signal



Spatial Frequency Domain Examples

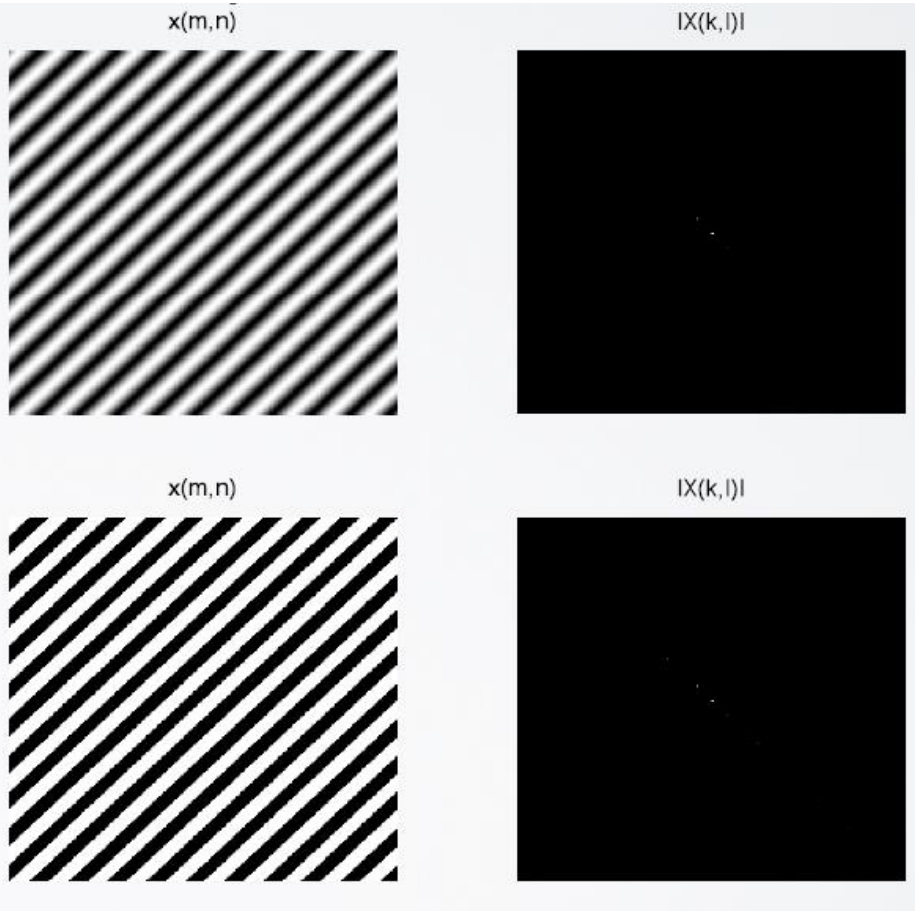
- Same direction, different frequencies





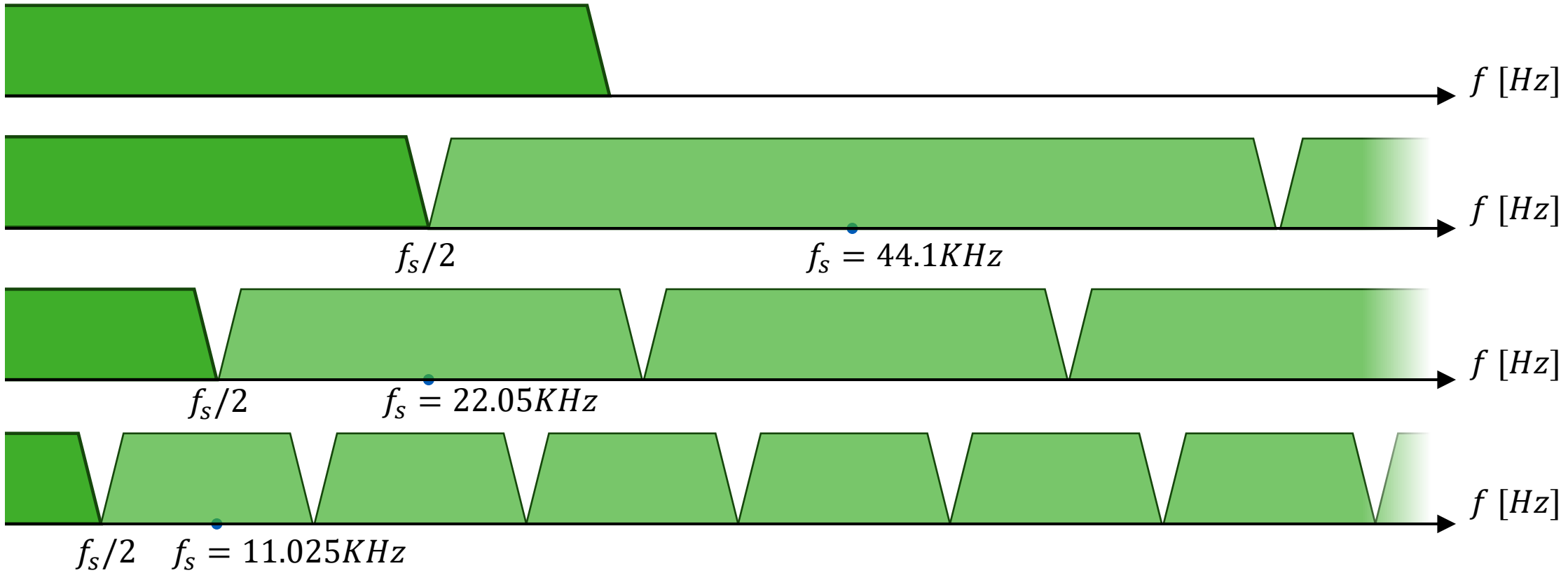
Spatial Frequency Domain Examples

- Same direction, same frequency but different edge sharpness



Audio sampling

- Short music sequence with $f_s = \{44.1KHz, 22.05KHz, 11.025KHz\}$ Demo



Summary

- A time continous signal is converted to time discrete form through sampling
- Fourier transform allows for seeing the spectral replication that occurs
- Multiplication in time (frequency) domain ↔ convolution in frequency (time) domain
- Perfect reconstruction back to continous time is possible if $f_s > f_{max}$
- Aliasing occurs if the Nyquist-Shannon sampling theorem is not followed

Links